

Product Requirements Document

EnterprateAI Credit-Based Usage System

Product: EnterprateAI

Feature: AI Credit Management and Subscription Usage Control

Version: 1.0

Priority: High

Prepared for: Development Team

Target release: Next production update

1. Product overview

EnterprateAI currently controls usage through fixed monthly feature allowances, such as:

- 5 idea validations
- 5 business plans
- 10 business proposals
- 30 sales letters
- 2 scenario simulations

This structure limits user flexibility and exposes the platform to unpredictable AI costs because users may consume every allowance independently.

The platform must be updated to use a unified AI credit system. Users will receive a defined number of credits based on their subscription plan and may use those credits across eligible AI-powered features.

Non-AI operational features such as customers, products, vendors, invoices and quotations will remain outside the credit system.

2. Objectives

The credit system must:

1. Give users flexibility to choose which AI outputs they generate.
2. Control and predict EnterprateAI's AI infrastructure costs.
3. Prevent excessive or abusive usage.
4. Support monthly subscriptions, annual subscriptions and free accounts.
5. Allow credit top-ups in the future.

6. Maintain a complete, auditable transaction history.
7. Deduct credits only after successful AI generation.
8. Allow administrators to change credit values without code changes.

3. Subscription plans

3.1 Explorer Plan

Price: £0

Billing type: Free

Credit allocation: 50 one-time AI credits

Credits do not refresh monthly.

The Explorer plan should include:

- 50 one-time AI credits
- Unlimited workspace and financial tools
- Unlimited products, customers and vendors
- Unlimited invoices and quotations
- Business Registration Guide
- 1 marketplace listing
- 1 user

The 50 credits should allow a user to generate approximately:

- 1 idea validation
- 1 business plan
- Limited additional regeneration or market intelligence

Explorer credits must be issued only once per verified account.

3.2 Starter Insight Plan

Monthly price: £19 per month

Annual price: £190 per year

Annual equivalent: £15.83 per month

Credit allocation: 500 AI credits per billing month

Starter Insight features:

- 500 AI credits every month
- Idea validation
- Business plan generation

- Business proposal generation
- Sales letter generation
- Scenario simulations
- Basic Fragility Index
- Adaptive Scenario Intelligence
- Unlimited products, customers and vendors
- Unlimited invoices and quotations
- 1 marketplace listing
- 1 user

Annual subscribers must receive 500 credits every month.

The system must not issue all 6,000 annual credits in advance.

4. Credit costs by feature

The initial credit allocation must be configured as follows:

AI feature	Credit cost
Idea validation	5 credits
Market data refresh	4 credits
Complete business plan	40 credits
Regenerate one business plan section	3 credits
Complete business proposal	25 credits
Regenerate one proposal section	2 credits
Complete sales letter	10 credits
Regenerate one sales letter section	1 credit
Scenario simulation	10 credits
AI Fragility Index interpretation	3 credits

These values must not be permanently hardcoded.

The administrator must be able to edit:

- Credit cost
- Feature name
- Feature status
- Minimum plan
- Whether a feature is credit-controlled
- Whether refunds apply after failed generation

5. Features that must not consume credits

The following actions must remain free:

- Creating or editing products
- Creating or editing customers
- Creating or editing vendors
- Creating invoices
- Creating quotations
- Editing generated documents manually
- Viewing existing documents
- Downloading existing documents
- Exporting existing documents
- Saving existing documents
- Using standard calculators
- Deterministic business scoring without AI narrative generation
- Accessing the Business Registration Guide
- Viewing marketplace listings
- Editing business profile information

Only actions that generate new AI content should consume credits.

6. Credit deduction rules

6.1 When credits are deducted

Credits must only be permanently deducted after the AI generation has completed successfully.

Recommended flow:

1. User initiates AI generation.
2. System checks available balance.
3. System places the required credits on hold.
4. AI request is processed.
5. If successful, held credits are committed.
6. If unsuccessful, held credits are released.

Credits must not be charged where:

- The AI provider returns an error.
- The request times out.

- The generation fails.
- The platform returns an invalid or empty result.
- The system crashes before the output is saved.
- A payment or permission error prevents completion.

6.2 Insufficient credits

When a user does not have enough credits:

- Do not start the AI request.
- Display the required credits.
- Display the current balance.
- Show the next credit reset date where applicable.
- Provide an upgrade or top-up option.

Example message:

This action requires 40 credits. You currently have 22 credits. Your credits reset on 1 August 2026. Upgrade or purchase additional credits to continue.

6.3 Duplicate submission protection

The platform must prevent multiple deductions caused by:

- Double-clicking a button
- Browser refresh
- Network retries
- Duplicate API requests
- Repeated webhook delivery

Every AI generation request must use a unique idempotency key.

Only one credit transaction may be completed for the same generation request.

7. Credit expiry and reset rules

7.1 Explorer credits

- 50 credits issued once.
- Credits do not reset.
- Credits remain valid while the account remains active.

- A user must not receive another free allocation by cancelling and recreating an account.

7.2 Starter monthly subscriptions

- 500 credits issued at the start of each billing month.
- Unused credits expire at the next monthly reset.
- Credits do not roll over.

7.3 Starter annual subscriptions

- 500 credits issued monthly.
- Credits reset on the monthly anniversary of the annual subscription start date.
- Unused credits expire at the next monthly reset.
- The system must not issue 6,000 credits at once.

7.4 Upgrade behaviour

When an Explorer user upgrades to Starter Insight:

- Existing Explorer promotional credits may remain available until used or expire after 30 days.
- Starter credits must be issued immediately after successful payment.
- Starter benefits begin immediately.

7.5 Downgrade behaviour

When a paid subscription expires or is cancelled:

- Paid-plan access remains active until the end of the paid billing period.
- Unused paid-plan credits expire at the end of that billing period.
- The account returns to Explorer.
- No additional Explorer credits should be issued if the free allocation was previously received.

8. User interface requirements

8.1 Credit balance display

The user's current credit balance must be visible in:

- Main dashboard header
- Account menu
- Subscription page
- AI generation screens

- Mobile navigation

Display format:

342 AI credits remaining

The display should include:

- Current balance
- Total monthly allocation
- Next reset date
- Link to usage history
- Upgrade or top-up button

Example:

342 of 500 credits remaining
Resets 1 August 2026

8.2 Credit confirmation before generation

Before an AI action is submitted, the interface must show its cost.

Examples:

Generate Business Plan — 40 credits

Run Scenario Simulation — 10 credits

Regenerate Section — 3 credits

Where appropriate, show a confirmation message:

This action will use 40 credits. Continue?

The user should have the option to disable confirmation for low-cost actions through account preferences.

8.3 Pricing page update

Remove the current fixed AI generation allowances from the Starter Insight card.

Replace them with:

Explorer

£0

- 50 one-time AI credits

- Generate an idea validation and business plan
- Unlimited workspace and financial tools
- Unlimited products, customers and vendors
- Unlimited invoices and quotations
- Business Registration Guide
- 1 marketplace listing
- 1 user

Starter Insight

£19/month

£190/year when billed annually

- 500 AI credits every month
- Use credits across idea validations, business plans, proposals, sales letters and simulations
- Basic Fragility Index
- Adaptive Scenario Intelligence
- Unlimited products, customers and vendors
- Unlimited invoices and quotations
- 1 marketplace listing
- 1 user

Add an information tooltip:

AI credits are used when generating new AI-powered outputs. Viewing, editing and downloading existing documents does not consume credits.

8.4 Usage history page

Create an AI Credit Usage page containing:

Date	Feature	Action	Credits	Status	Balance
11 July 2026	Business Plan	Full generation	-40	Completed	460
11 July 2026	Idea Validation	Validation	-5	Completed	455
12 July 2026	Business Plan	Section regeneration	-3	Completed	452

The user must be able to filter by:

- Date

- Feature
- Transaction type
- Status
- Credit deduction
- Credit addition

9. Credit transaction types

The ledger must support:

- Initial allocation
- Monthly allocation
- Subscription renewal
- Promotional credit
- AI generation deduction
- Credit hold
- Credit hold release
- Failed-generation refund
- Manual admin adjustment
- Top-up purchase
- Subscription upgrade
- Subscription downgrade
- Expiry
- Chargeback adjustment

Every transaction must include:

- User ID
- Workspace ID
- Subscription ID
- Feature code
- Document or generation ID
- Credit amount
- Balance before
- Balance after
- Transaction type
- Transaction status
- Timestamp
- Idempotency key
- Reason or description

- Administrator ID where applicable

10. Recommended data model

10.1 Credit wallets

credit_wallets

Field	Type
id	UUID
user_id	UUID
workspace_id	UUID
plan_id	UUID
available_credits	Integer
held_credits	Integer
lifetime_credits_issued	Integer
lifetime_credits_used	Integer
next_reset_at	Timestamp
last_reset_at	Timestamp
created_at	Timestamp
updated_at	Timestamp

10.2 Credit transactions

credit_transactions

Field	Type
id	UUID
wallet_id	UUID
user_id	UUID
workspace_id	UUID
feature_code	String
generation_id	UUID, nullable
transaction_type	Enum
credits	Integer
balance_before	Integer
balance_after	Integer
status	Enum
idempotency_key	String, unique

Field	Type
description	Text
expires_at	Timestamp, nullable
created_at	Timestamp

Suggested transaction types:

- allocation
- deduction
- hold
- hold_release
- refund
- expiry
- top_up
- admin_adjustment
- promotion

Suggested statuses:

- pending
- completed
- failed
- reversed

10.3 Feature credit configuration

credit_feature_config

Field	Type
id	UUID
feature_code	String, unique
feature_name	String
credit_cost	Integer
enabled	Boolean
minimum_plan	String
refundable_on_failure	Boolean
created_at	Timestamp
updated_at	Timestamp

Example feature codes:

- idea_validation
- market_data_refresh
- business_plan_full

- business_plan_section
- proposal_full
- proposal_section
- sales_letter_full
- sales_letter_section
- scenario_simulation
- fragility_ai_interpretation

10.4 Subscription credit configuration

plan_credit_config

Field	Type
id	UUID
plan_code	String
allocation_type	Enum
credits_per_period	Integer
reset_frequency	Enum
rollover_enabled	Boolean
expiry_enabled	Boolean
enabled	Boolean
created_at	Timestamp
updated_at	Timestamp

Suggested allocation types:

- one_time
- monthly
- promotional
- top_up

11. API requirements

11.1 Get credit balance

GET /api/credits/balance

Example response:

```
{
  "plan": "starter_insight",
  "available_credits": 342,
  "held_credits": 0,
```

```
"monthly_allocation": 500,  
"next_reset_at": "2026-08-01T00:00:00Z"  
}
```

11.2 Get feature cost

GET /api/credits/features

Example response:

```
{  
  "features": [  
    {  
      "feature_code": "idea_validation",  
      "feature_name": "Idea Validation",  
      "credit_cost": 5  
    },  
    {  
      "feature_code": "business_plan_full",  
      "feature_name": "Complete Business Plan",  
      "credit_cost": 40  
    }  
  ]  
}
```

11.3 Check credit availability

POST /api/credits/check

Request:

```
{  
  "feature_code": "business_plan_full"  
}
```

Response:

```
{  
  "allowed": true,  
  "required_credits": 40,  
  "available_credits": 342,  
  "remaining_after_action": 302  
}
```

11.4 Reserve credits

POST /api/credits/reserve

Request:

```
{
  "feature_code": "business_plan_full",
  "generation_id": "generation_uuid",
  "idempotency_key": "unique_request_key"
}
```

11.5 Commit credits

POST /api/credits/commit

Request:

```
{
  "generation_id": "generation_uuid",
  "idempotency_key": "unique_request_key"
}
```

11.6 Release reserved credits

POST /api/credits/release

Used where generation fails.

11.7 Get usage history

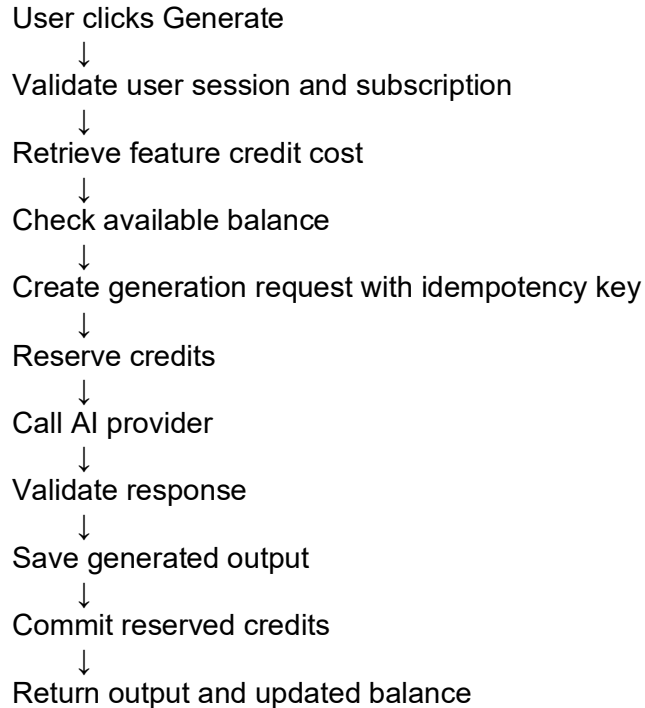
GET /api/credits/transactions

Supported filters:

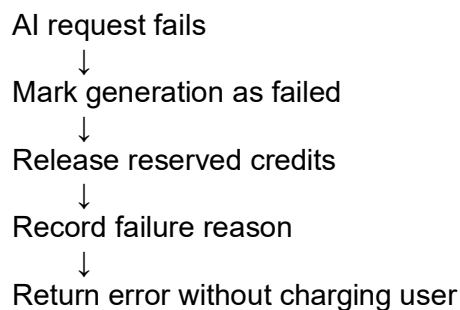
- Start date
- End date
- Feature code
- Transaction type
- Status
- Pagination

12. AI generation workflow

The required generation sequence is:



Failure sequence:



13. Business plan generation optimisation

The current AI usage report indicates that full-document generation and section generation may be happening within the same workflow.

The development team must review the generation architecture.

Preferred approach:

1. Generate the full document once as structured JSON.
2. Store every section separately.
3. Render the complete document from stored sections.
4. Call the AI again only when the user deliberately regenerates a specific section.
5. Deduct section-regeneration credits only for that selected section.

The platform must not automatically generate both a complete document and every individual section as separate AI calls unless technically necessary and approved.

14. Top-up credits

Top-up purchases should be supported either in this release or prepared for a second phase.

Suggested packages:

Package	Price
100 AI credits	£4.99
300 AI credits	£11.99

Top-up credits should:

- Be added immediately after confirmed payment.
- Remain available until consumed.
- Not expire at the monthly subscription reset.
- Be consumed after subscription credits where technically practical.
- Be included in the same wallet balance while retaining their source in the ledger.

Top-ups should be available to paid subscribers initially.

Explorer users should be prompted to upgrade before purchasing top-ups unless otherwise approved.

15. Stripe billing requirements

The credit system must integrate with existing Stripe subscriptions.

Required Stripe events:

- checkout.session.completed
- invoice.paid
- invoice.payment_failed
- customer.subscription.created
- customer.subscription.updated
- customer.subscription.deleted
- charge.refunded
- charge.dispute.created

Required actions:

Monthly subscription payment succeeds

- Confirm Starter Insight status.
- Issue or reset balance to 500 monthly credits.
- Set next reset date.

Annual subscription payment succeeds

- Activate annual Starter Insight.
- Issue 500 credits.
- Schedule monthly credit resets based on subscription anniversary.

Payment fails

- Do not issue new credits.
- Apply grace-period rules.
- Notify the user.

Subscription cancelled

- Preserve access until the paid period ends.
- Stop future credit allocations.
- Downgrade to Explorer at period end.

All Stripe webhook processing must be idempotent.

16. Admin requirements

The admin dashboard must allow authorised staff to:

- View a user's credit balance
- View credit transaction history
- Add credits
- Remove credits
- Refund credits
- Edit credit costs by feature
- Enable or disable credit-controlled features
- View high-usage accounts
- View failed generations
- View AI cost per feature
- View credits used per feature
- Export credit usage reports
- Search by user email or workspace

Manual adjustments must require:

- Administrator identity
- Reason
- Number of credits
- Timestamp
- Audit log

17. Analytics and reporting

The system must track:

- Credits issued
- Credits used
- Credits expired
- Credits refunded
- Average credits used per user
- Average credits used by plan
- Feature-level credit consumption
- Actual AI cost per feature
- Revenue per active subscriber
- AI gross margin per subscriber
- Number of failed generations
- Number of regenerated sections
- Number of users who exhaust their credits
- Upgrade conversion after insufficient-credit prompts
- Top-up conversion rate

Recommended internal dashboard metrics:

Metric	Description
Credit utilisation rate	$\text{Credits used} \div \text{credits allocated}$
AI cost per active user	$\text{Total AI cost} \div \text{active users}$
Revenue-to-AI-cost ratio	$\text{Subscription revenue} \div \text{AI cost}$
Feature cost variance	Actual cost compared with assigned credit cost
Credit exhaustion rate	Users reaching zero balance
Regeneration rate	Repeated generation requests per document

18. Migration requirements

Existing users must be migrated without disruption.

Explorer users

- Issue 50 one-time credits.
- Mark the free allocation as already issued.
- Preserve existing data and documents.

Starter Insight monthly users

- Replace existing feature quotas with 500 credits.
- Issue 500 credits on migration.
- Set the next reset date to the user's next billing date.

Starter Insight annual users

- Issue 500 credits on migration.
- Set monthly resets based on the annual subscription start date.
- Do not issue the full annual allocation.

Existing generated documents must remain accessible and must not require credits to open, edit or download.

19. Security and abuse protection

The system must include:

- Email verification before Explorer credits are issued
- Rate limiting on AI endpoints
- Unique account and workspace checks
- Idempotent credit deductions
- Fraud detection for repeated free account creation
- Server-side credit validation
- No client-side-only balance enforcement
- Complete transaction audit trail
- Role-based admin access
- Protection against negative balances
- Database locking or atomic transactions during deduction

Credit balance updates must use atomic database operations.

The balance must never become negative.

20. Error handling

Recommended error codes:

Code	Meaning
INSUFFICIENT_CREDITS	User does not have enough credits
FEATURE_DISABLED	Feature is not currently available
PLAN_NOT_ELIGIBLE	User's plan cannot access the feature
CREDIT_RESERVATION_FAILED	Credits could not be reserved
GENERATION_FAILED	AI generation failed
DUPLICATE_REQUEST	Same idempotency key already processed
SUBSCRIPTION_INACTIVE	Paid subscription is inactive
CREDIT_CONFIGURATION_MISSING	Feature cost has not been configured

Errors shown to the user must be written in clear language and must not expose internal technical information.

21. Acceptance criteria

The update will be accepted when:

1. Explorer users receive exactly 50 one-time credits.
2. Starter users receive 500 credits per monthly cycle.
3. Annual users receive 500 credits monthly rather than 6,000 upfront.
4. Credit costs are retrieved from configurable backend records.
5. The correct number of credits is displayed before each AI action.
6. Credits are deducted only after successful generation.
7. Failed generations automatically release or refund credits.
8. Duplicate requests do not produce duplicate deductions.
9. Users can view their balance and transaction history.
10. Admin users can adjust balances and feature costs.
11. Existing documents remain accessible without credit charges.
12. Monthly credits expire and reset correctly.
13. Top-up credits, when enabled, are added after confirmed payment.
14. Stripe renewal events trigger the correct credit allocation.
15. The database prevents a negative credit balance.
16. Usage analytics are available by user, plan and feature.
17. The pricing page reflects the new credit-based structure.

- 18. Mobile and desktop interfaces show the same accurate balance.
- 19. Full-document and section generation are not duplicated unnecessarily.
- 20. All credit transactions are auditable.

22. Out of scope for the first release

The following may be delivered later:

- Credit gifting between users
- Credit transfers between workspaces
- Team-shared credit pools
- Enterprise custom credit contracts
- Automatic credit rollover
- Credit rewards for referrals
- Promotional campaigns with expiring credits
- Country-specific credit pricing
- Real-time variable credit pricing based on token use

23. Recommended delivery phases

Phase 1: Core credit system

- Credit wallets
- Credit ledger
- Plan allocations
- Feature credit costs
- Balance checks
- Credit reservation and commitment
- Explorer and Starter plan migration
- User balance display

Phase 2: Billing and management

- Stripe renewal integration
- Annual monthly resets
- Usage history
- Admin controls
- Analytics
- Failed-generation refunds

Phase 3: Commercial expansion

- Credit top-ups
- Promotional credits
- Usage alerts
- Credit exhaustion campaigns
- Advanced AI cost and margin reporting

24. Final product requirement

The credit system must be flexible, transparent and auditable.

Users must understand:

- How many credits they have
- How many credits an action costs
- When credits reset
- Why credits were deducted
- What happens when credits are exhausted

The platform must retain full control of credit pricing, feature availability and subscription allocations without requiring code changes for routine commercial adjustments.